Surface Ion Trap Optimization

Geometry, stability and fabrication

A numerical investigation of a single calcium-ion surface trap

Design development • Simulation findings • Material comparison

Source: Project investigation: working notebook and annotated decision record

Surface ion trap: findings and recommendation

151.75 meV predicted RF-only depth

Symmetric 200 µm RF rails • 5 µm gaps
5 µm gold • 500 µm fused silica

100 V peak | 50.141 MHz | q = 0.395

Source: Project finite-gap thickness screen: thickness_screen_results.json

Objective and constraints

Constraint	Limit
Height / radial frequency	50–100 μm / ≥1 MHz
Mathieu / voltage	q ≤0.4 / ≤100 V peak
RF width / gap	20–200 μm / 5–20 μm
Drive frequency	10–60 MHz

Extended design: center width and layer thicknesses may vary.

Source: starter_file_given.ipynb — parameter and score cells

Source: Working notebook — revised center and metal thickness

Notebook development: three key steps

- Start: lower height and lower RF frequency to increase depth.
- Failure: low-frequency trials exceeded the Mathieu limit.
- Revision: correct the 3.68 ratio and retune frequency near q = 0.4.

Recorded milestone: ~150.5 meV → current saved output: 151.622 meV.

Source: Traped ion working file.ipynb — notes in cells 3–9; saved outputs 20 and 24

Why the final ratio differs from 3.68

Depth □ V^2/f^2 q □ V G/f^2

- 3.68 means RF width / RF separation—not fabrication gap.
- The paper holds height and RF settings fixed; the constrained search varies them.
- Selected effective ratio: (200 + 5)/(26.71361 + 5) ≈ 6.46.

Source: [1] Nizamani & Hensinger, Appl. Phys. B (2012), doi:10.1007/s00340-011-4803-x

Source: Project constrained analytic search: optimize_trap.py

Symmetric rails led the depth search

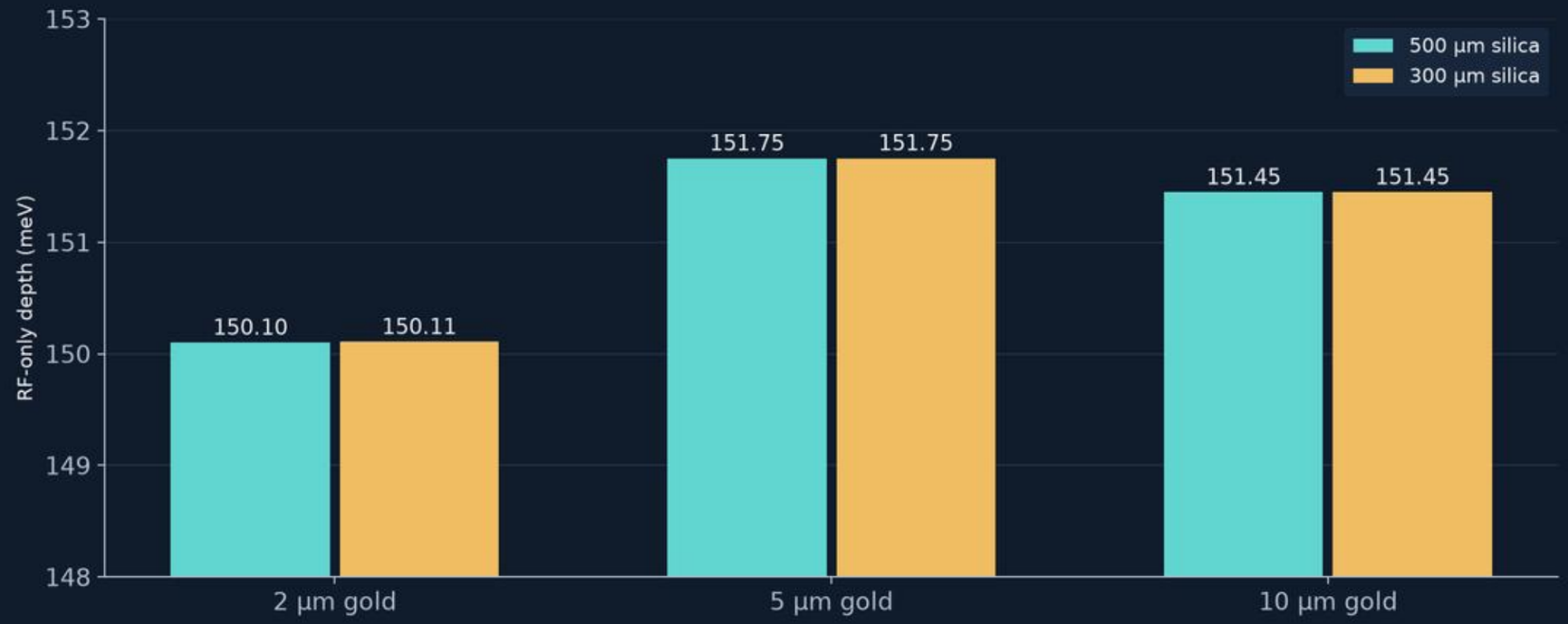
RF rails; center =60 µm	Predicted depth
Equal: 200 / 200 µm	144.29 meV
Independent-width search: ≈200 / 200 µm	144.29 meV
Illustrative unequal: 200 / 100 µm	134.20 meV

Asymmetry can aid mode orientation; it did not improve this modeled depth objective.

Source: Project gapless-plane results: trap_optimization_results.json

Source: [2] Allcock et al., New J. Phys. (2010), doi:10.1088/1367-2630/12/5/053026

Thickness: retain 5 µm gold on 500 µm silica



Wafer thinning offered negligible benefit; the 5 vs. 10 µm ranking needs a finer mesh.

Source: Project six-case finite-gap field results; retuned to q =0.395

Source: [3] Britton, microfabrication dissertation (2008) — plated-gold fabrication precedent

Cu/Au may reduce loss—not ideal trap depth

~23% lower estimated electrode resistance

- Candidate: 4.8 µm copper + 0.2 µm gold; same 5 µm total thickness.
- At the same 100 V electrode amplitude: no demonstrated depth gain.
- Film quality, coating integrity and RF feed losses require validation.

Source: [6] Tao et al., working Cu/Au surface trap, arXiv:2002.11470 (2020)

Source: [7] Matula, J. Phys. Chem. Ref. Data (1979); resistance estimates in rf_material_comparison.json

Final candidate and saved notebook result

Parameter	Recommended screened candidate
RF width / gap / center	200 / 5 / 26.71361 μm
Gold / silica thickness	5 / 500 μm
Voltage / frequency / q	100 V / 50.141 MHz / 0.395
Depth / height / radial minimum	151.75 meV / 58 μm / 6.945 MHz

Notebook saves 151.622 meV at 49.4 MHz; domain and q differ, so compare cautiously.

Source: Fixed-window field screen + frequency scaling: recommended_thickness_candidate.json

Source: Current saved notebook outputs — inspected, not re-executed for this deck

What remains before fabrication

- Verify the final drive point with mesh and domain convergence.
- Include axial DC confinement and the full escape barrier.
- Check finished gaps, adhesion, post-bake coating and breakdown.
- Measure the RF network, voltage balance, power loss and temperature.

Source: [5] Schmied, finite-electrode and dielectric corrections, New J. Phys. (2010)

Source: [11] Copper diffusion through gold coatings, Anal. Chim. Acta (2023)